

SignalScope — one-page model report

SIH 2026 internal, Problem Statement 2 · model **mixed_resnet18_native_v1_calibrated** · SHA-256 eab7d9d82c8250bd... · 2026-09-12 · github.com/sibtainmunshi/Signal_Scope

Self-evaluated public-data results. Organizers supplied no dataset, baseline or hidden-test scores; those fields are **not supplied**, not estimated.

Task and scope

Binary real vs AI-generated image classification with a continuous AI-positive score; intended for general objects, scenes and products. Built: core detector, A (model-linked explanation + audit), C (robustness), D (EXIF only; C2PA not checked), F (local web app/CLI), G (bounded post-processing attack analysis).

Data and splits

Source	Use	Real / AI
CIFAKE (CIFAR-10 vs SD1.4, 32 px)	train 8,000 of 79,689	4,000 / 4,000
CIFAKE grouped val / calibration	selection / calibration	5,000 / 4,964 val; 5,000 / 4,969 calibration
GenImage BigGAN + SD1.5 (train folders)	train 6,239	3,133 / 3,106
GenImage val / calibration	selection / calibration	369 / 414; 397 / 384
UniversalFakeDetect dev	evaluation only	guided vs ImageNet, LDM vs LAION, 500+500 each
Reserved final	scored once after freeze	4,000 AI + 500 shared LAION real; CIFAKE test 10k + 10k

Exact-overlap and perceptual-hash screening against protected data; grouped splits. Hash screening is not exhaustive. Audit found real JPEG vs generated square PNG in both GenImage and the external set, so training balances JPEG and every external result is also reported **format-matched** (identical crop, resize and JPEG for both labels).

Model and training

- ImageNet-pretrained ResNet-18, fine-tuned end to end (AdamW 2e-4, cosine, 10 epochs, batch 64, AMP); epoch chosen by mean GenImage/CIFAKE val AUC.
- Random 128 px crops; small images upscaled, BigGAN real training photos resized to 128 px. GenImage generated training images receive JPEG compression; symmetric rescale, blur, flip and crop-JPEG augmentation.
- Inference: mean logit of up to five native 128 px crops; CPU ~15–50 ms per image on our laptop.
- Temperature 1.65 fit on calibration splits; threshold 0.455 = strictest validation threshold with real FPR ≤ 5.0% per domain. ECE CIFAKE 0.027→0.009, GenImage 0.047→0.061 (worse).

Explanation (Module A)

Returned-class Grad-CAM per crop, stitched, plus a masking diagnostic. 40 fixed dev images (27 correct): masking the top Grad-CAM window lowered the verdict score more than random windows in 72.5% (mean fill, p=0.002) and 67.5% (blur); effects are small. JPEG map stability median rho 0.95; weight-randomization rho 0.52 (maps only partly weight-dependent). No named artifact (text, lighting, anatomy) is claimed.

Results at the frozen operating point (threshold 0.455)

Evaluation	ROC-AUC [95% CI]	Macro-F1	Accuracy	Real FPR	AI TPR
CIFAKE validation (in-domain)	0.993	0.951	95.2%	2.2%	92.5%
GenImage BigGAN/SD1.5 validation	0.956	0.898	89.8%	4.9%	85.0%
External dev, unseen generators	0.647 [0.624, 0.670]	0.513	57.8%	8.4%	23.9%
External dev, format-matched	0.653 [0.628, 0.677]	0.558	59.2%	13.9%	32.4%
Reserved GLIDE/DALLE (final)	0.565	0.400	42.1%	10.4%	18.3%
Reserved, format-matched (final)	0.643	0.472	47.9%	14.6%	29.2%
CIFAKE author test 20k (final)	0.993	0.951	95.1%	1.9%	92.2%

External per-generator AUC (as distributed; then matched): guided 0.630; Idm_200 0.665; guided 0.611; Idm_200 0.695. External rates are means over generator pairs. Reserved pairs contain 1,000 AI + 500 shared real images; GLIDE x3 and DALLE represent two families. Development CIs condition on these images and model selection.

CIFAKE author test confusion matrix

	Pred. real	Pred. AI
Real	9808	192
AI	783	9217

Baseline comparison (unseen-generator mean AUC, as distributed / matched)

Organizer baseline	not supplied
Our CIFAKE-only ResNet-18 (v0.1.0)	dev 0.553 / 0.540; reserved 0.548 / 0.546; dev real FPR 84.3%
Selected model	dev 0.647 / 0.653; reserved 0.565 / 0.643; dev real FPR 8.4%

Seven candidates were compared on the same development data (including a frozen CLIP head whose 0.708 as-distributed AUC fell to 0.542 when format-matched); reserved data were never used for selection.

Robustness (Module C/G)

GenImage val AUC original 0.956; JPEG q70 0.953; q30 0.906; half resolution 0.896; blur 0.937. Six-transformation search flipped 19.8% of initially correct predictions; on 32 px CIFAKE, half resolution drops AUC to 0.726 (flip rate 42.6%).

Limitations and failure cases

- Unseen-generator AUC is modest. On the reserved unseen set the mean AUC is 0.565 / 0.643 (as distributed / format-matched); GLIDE is weakest. At the conservative threshold most unseen AI images are missed.
- PNG outputs scored worse than matched JPEG inputs here. Matching changes both geometry and encoding, so this does not isolate a causal file-format effect.
- Calibration is domain-specific: confidence and FPR need not transfer to new generators or pipelines.
- Resizing, blur and strong JPEG raise real-image false positives; tiny upscaled images are fragile.
- Grad-CAM is model attribution, not verified defects. Audit sample is small; two-person usefulness review is pending.
- GenImage-derived weights are for noncommercial research use (CC BY-NC-SA 4.0).